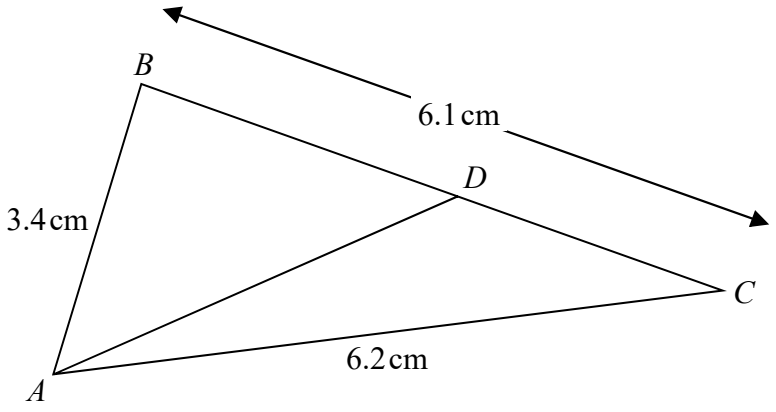
18 The diagram shows triangle ABC.



$AB = 3.4\,\text{cm}$ $AC = 6.2\,\text{cm}$ $BC = 6.1\,\text{cm}$

D is the point on BC such that

$$\text{size of angle } DAC = \frac{2}{5} \times \text{ size of angle } BCA$$

Calculate the length DC.
Give your answer correct to 3 significant figures.
You must show all your working.

Cosine rule: $\cos C = (6.1^2+6.2^2-3.4^2)/(2\times6.1\times6.2)$.

So angle BCA $\approx 32.0805°$.

Angle DAC $= (2/5)\times32.0805° \approx 12.8322°$.

Angle ADC $= 180° - 32.0805° - 12.8322°$

$\approx 135.0873°$. By sine rule,

$DC/\sin12.8322° = 6.2/\sin135.0873°$.

Therefore DC ≈ 1.95 cm (3 s.f.).

5 marks total
Cosine-rule angle
Angle DAC
Sine-rule setup
Complete DC calculation
1.95 cm

..... cm

(Total for Question 18 is 5 marks)